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Unit – 5 : Session – 8 : SLO – 2 | ARM CPSR, Modes and Software Interrupts

■ I. Fill in the Blanks

1. MRS — MRS Rd, CPSR moves the status register (including mode bits) into a general register.
2. Supervisor (SVC) — SWI triggers a software interrupt that enters Supervisor mode.
3. R8 — FIQ mode banks registers R8–R14.

■ II. True or False

1. TRUE — FIQ banks R8–R14 (7 registers) vs IRQ's R13–R14 (2), so FIQ handlers avoid stacking overhead.
2. FALSE — the CONTROL register belongs to ARMv7-M (Cortex-M); ARMv7-A switches modes via CPSR writes/exceptions.
3. TRUE — SWI is precisely the User → Supervisor transition mechanism (system calls).

■ III. Match the Following

1. SWI — 3. Software interrupt.
2. CPSR — 2. Holds processor status.
3. FIQ — 1. Fast interrupt mode.

■ IV. Numerical Answers

1. 5 — CPSR mode bits M[4:0] determine the processor mode.
2. 7 — FIQ banks R8_fiq through R14_fiq.
3. 2 — IRQ banks only R13_irq and R14_irq.